

Muhammad Fakhrrur Rozi

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📄 rozy.my.id | 📝 blog.rozy.my.id | 📍 Central Java, Indonesia

Self-taught in CS during an Arabic Literature degree – driven by fascination with formal languages and type systems

Education

Sarjana Sastra Arab (Arabic Literature)

MA'HAD ALY ANDALUSIA BANYUMAS

2021 - 2025

Banyumas, Indonesia

- GPA: 3.65/4.0
- Self-taught in CS during and after this degree, driven by fascination with formal languages and type systems

High School

SMA ISLAM ANDALUSIA KEBASEN

2017 - 2020

Banyumas, Indonesia

Experience

Secretary / Administrative Automation

YAYASAN AL ANWAR AL HISYAMIYYAH – PONDOK PESANTREN

2020 - Present

Banyumas, Indonesia

- Built Playwright automation scripts to batch-enter student data for 3,000+ santri into the EMIS (Education Management Information System)
- Developed Node.js docx generators and OOXML manipulation tools to produce Laporan Pertanggungjawaban (accountability reports) for Pondok management
- Designed document automation workflows combining Playwright, OOXML, and custom scripts across 10+ report types

Programming Instructor

MA'HAD ALY ANDALUSIA (EXTRACURRICULAR)

First half 2026

Banyumas, Indonesia

- Taught programming fundamentals to 10+ students with no prior coding experience over 6 months
- Designed curriculum and hands-on exercises from scratch for absolute beginners
- Developed teaching materials and guided students through practical coding sessions

Projects

derive – Koka Derive CLI

GITHUB.COM/LITFILL/DERIVE

- CLI tool eliminating boilerplate by auto-deriving **show**, **eq**, and typeclass-like functions in Koka (2 derivation kinds, file I/O, dry-run mode)
- Built because Koka lacked metaprogramming – solved a real workflow pain point for Koka developers

Koka CLI Metaprogramming

sina – Hindley-Milner Type Inference

GITHUB.COM/LITFILL/SINA

- Full HM type inference (Algorithm W) from scratch in Haskell: parser, AST, constraint-based type checker
- Supports integers, booleans, conditionals, let-bindings, and binary ops with let-polymorphism

Haskell Type Systems PLT

klens – Value-based Lenses in Koka

GITHUB.COM/LITFILL/KLENS

- Value-based lenses for Koka: one-shot focused views with **set**, effectful **modify** propagating Koka effects, and composition
- Tuple optics and identity lens – a design exercising Koka's effect system

Koka Optics Effects

src-todo – TODO Tracker CLI

GITHUB.COM/LITFILL/SRC-TODO

- Haskell CLI managing TODO comments in source files: register, list, and unregister todos with unique UUIDs
- In-place file updates; comment parsing built with Megaparsec

Haskell CLI Megaparsec

Open Source & Certifications

2025 **BNFC (BNF Converter) – Bug fix**, github.com/BNFC/bnfc
English Language Certification, BLK, Kementrian Ketenagakerjaan
Noctalia Shell – Custom Plugins, github.com/LitFill/noctalia-plugins

Remote
Indonesia
Remote

Skills

- Primary Languages

●●●●● Haskell | Koka | OCaml | Rust | TypeScript/JavaScript
- Familiar With

Go | Idris2 | Agda | Lean | Common Lisp | Nushell
- Haskell Ecosystem

●●●●○ servant | warp | aeson | megaparsec | lens | QuickCheck | hedgehog | streaming | polysemy | hspec | tasty | req | hasql | relude
- Tools

Git | Linux (NixOS, Arch) | Neovim | Cabal | GHCup | HLS | Stack | Nix | hledger | Node.js | Playwright
- Soft Skills

Self-directed learning (15+ languages independently) | Technical communication | Teaching & mentoring